

Fragrance DNA: A Graph-Based Analysis of Scent Profiles, Note Communities, and Recommendation

EECS4414: Final Project Report

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Abstract

Fragrance discovery is challenging because perfumes are described through subjective combinations of notes, accords, brands, ratings, and popularity. Rating- and popularity-based recommendation can identify well-liked perfumes, but they often provide limited explanation for why two fragrances are similar. This project presents *Fragrance DNA*, a graph-based approach for analyzing scent profiles and generating explainable perfume recommendations. Using a FrAGRantica-derived dataset, we construct perfume-note, note co-occurrence, and perfume similarity graphs. We analyze central scent notes, detect note communities, and evaluate graph-based recommendations against rating-based, popularity-based, same-brand, and raw note-overlap baselines. The results show that the graph-based method substantially outperforms rating, popularity, and brand-only baselines on scent similarity while remaining competitive with direct note overlap. Overall, the project demonstrates that graph structure can provide an interpretable representation of fragrance relationships and support explainable perfume recommendation.

Keywords

fragrance, perfume recommendation, network analytics, graph mining, community detection, similarity networks

1 Introduction

Fragrance discovery is difficult because perfume descriptions are subjective and are usually based on combinations of notes, accords, brands, ratings, and popularity. Two perfumes may smell related even if they differ in brand or rating, while two highly rated perfumes may not be similar in scent profile. This makes rating-based recommendation useful for popularity ranking, but limited for explaining fragrance similarity.

This project, *Fragrance DNA: A Graph-Based Analysis of Scent Profiles, Note Communities, and Recommendation*, studies perfume data using graph-based methods. We model perfumes, notes, and accords as information networks in order to expose direct and indirect fragrance relationships. The project focuses on three main tasks: identifying central scent components, detecting interpretable

note communities, and producing explainable perfume recommendations based on graph similarity.

The evaluation shows that the graph-based method performs better than rating-based, popularity-based, and same-brand baselines on scent similarity, while remaining close to raw note-overlap. Unlike simple ranking strategies, the graph-based approach provides interpretable evidence through shared notes, shared accords, central scent components, and graph-based relationships.

1.1 Related Work

Recommendation systems commonly use collaborative filtering, content-based filtering, item similarity, and ranking methods to suggest relevant items [1]. Graph-based recommendation extends these ideas by modeling relationships among entities, especially in user-item and item-feature networks [2]. Network analytics provides measures such as degree, centrality, Jaccard similarity, cosine similarity, and clustering for comparing entities and identifying important nodes [3, 5, 7]. Community detection methods such as Louvain identify clusters by optimizing modularity [4]. This project applies these methods to fragrance data from FrAGRantica [8], representing perfumes, notes, and accords as graph structures to discover scent communities, identify influential notes, and generate explainable recommendations.

2 Problem Definition

Let P be the set of perfumes, N the set of scent notes, and A the set of accords. Each perfume $p \in P$ is represented by a set of notes $N_p \subseteq N$ and a set of accords $A_p \subseteq A$. The goal of the project is to construct graph representations that preserve these relationships and use them to analyze fragrance structure. Specifically, we seek to identify structurally important notes, detect interpretable note communities, and recommend perfumes that are similar to a query fragrance.

For recommendation, the input is a query perfume $q \in P$ and the output is a ranked list of top- k perfumes $R_q = \{p_1, \dots, p_k\}$ that are similar to q according to scent-profile and graph-based evidence. A useful recommendation should share meaningful notes or accords with the query perfume, be close to it in the perfume similarity graph, and remain explainable through observable scent attributes.

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The project therefore addresses three connected network-analysis questions. First, which scent notes are structurally central in the fragrance graph? Second, do groups of notes form communities that correspond to interpretable scent families or fragrance styles? Third, does graph-based ranking produce more scent-similar recommendations than simple rating-, popularity-, brand-, or note-overlap baselines?

3 Dataset and Preprocessing

The project uses a Kaggle dataset derived from *Fragrantica* fragrance records. The dataset includes perfume metadata, ratings, top, middle, and base notes, source URLs, and up to five main accords. These fields directly support the planned graph structures: perfumes as item nodes, notes as feature nodes, and accords as additional scent-profile attributes.

The cleaned source file contains 24,063 perfume records with unique URLs. It includes 1,060 unique brands, 54 countries, 1,671 unique notes, and 84 unique accords. Preprocessing converted rating values from comma-based decimals to numeric decimal format, split top, middle, and base note fields into perfume-note edge rows, and converted accord columns into perfume-accord edge rows while preserving accord rank.

The processed dataset consists of three graph-ready files: a perfume metadata table with 24,063 records, a perfume-note edge list with 238,357 edges, and a perfume-accord edge list with 118,819 edges. These files provide the input for graph construction, similarity computation, recommendation, evaluation, and visualization.

Table 1: Processed dataset summary.

Property	Count
Perfumes	24,063
Brands	1,060
Countries	54
Notes	1,671
Accords	84
Perfume-note edges	238,357
Perfume-accord edges	118,819

The main dataset limitations are user-generated note and accord labels, missing release year or perfumer fields for some records, and the absence of user-level interaction histories. As a result, recommendation evaluation relies on item metadata, note overlap, accord similarity, rating information, and qualitative case studies rather than personalized user-behaviour data.

4 Graph Construction and Methodology

The processed files were used to construct three graph structures: a perfume-note bipartite graph, a note co-occurrence graph, and a perfume similarity graph. Together, these graphs support note analysis, community detection, perfume similarity, and recommendation. Figure 1 summarizes this workflow.

4.1 Network Construction

The first graph is the perfume-note bipartite graph $G_B = (P \cup N, E_B)$, where P represents perfume nodes, N represents note nodes, and

each edge (p, n) indicates that note n appears in perfume p . The edge list preserves the note layer as top, middle, or base, which keeps information about whether a note appears in the opening, heart, or base of a fragrance. The bipartite graph contains slightly fewer edges than the processed perfume-note table because duplicate or invalid graph edges were removed during graph construction.

The second graph is the note co-occurrence graph $G_N = (N, E_N)$. In this graph, two notes are connected when they appear in at least one shared perfume. Edge weights are computed using Jaccard similarity over shared perfumes:

$$w(n_i, n_j) = \frac{|P_{n_i} \cap P_{n_j}|}{|P_{n_i} \cup P_{n_j}|}. \quad (1)$$

This normalization reduces the effect of very common notes such as musk, amber, vanilla, or bergamot, which may otherwise dominate the graph only because they appear frequently.

The third graph is the perfume similarity graph $G_P = (P, E_P)$. In this graph, perfumes are connected based on similarity over their note and accord profiles. A similarity threshold of $\tau = 0.3$ was used to remove weak edges and keep the graph more interpretable than a fully dense similarity network.

Table 2: Constructed graph summary.

Graph	Nodes	Edges	Density
G_B bipartite	25,734	236,985	0.000716
G_N note graph	1,671	107,434	0.076998
G_P perfume graph	24,063	1,524,828	0.005267

The note co-occurrence graph is connected, with one component containing all 1,671 note nodes. The perfume similarity graph contains 243 connected components. This suggests that most perfumes are connected through note or accord similarity, while some perfumes remain in smaller isolated groups. The density values also show that the three graphs serve different purposes: the bipartite graph is sparse because it records direct perfume-note membership, the note graph is denser because many notes co-occur across perfumes, and the perfume graph balances coverage with interpretability by thresholding weak similarities.

4.2 Similarity Measures and Recommendation

The recommendation pipeline produces top- k recommendations using the perfume similarity graph. For a query perfume, candidate perfumes are ranked using a composite score that combines three signals: Personalized PageRank, direct perfume-similarity edge weight, and a popularity prior. Personalized PageRank receives weight 0.5 and captures graph-based proximity from the query perfume. The direct edge weight receives weight 0.3 and is computed from a weighted blend of note Jaccard similarity, accord Jaccard similarity, and weighted note overlap. The popularity prior receives weight 0.2 and uses rating value and rating count to slightly prefer well-supported perfumes. The final ranking is therefore based primarily on graph proximity and scent-profile similarity, while still incorporating a smaller popularity signal.

The recommendation output supports explainability by reporting shared notes, shared accords, and similarity evidence for each

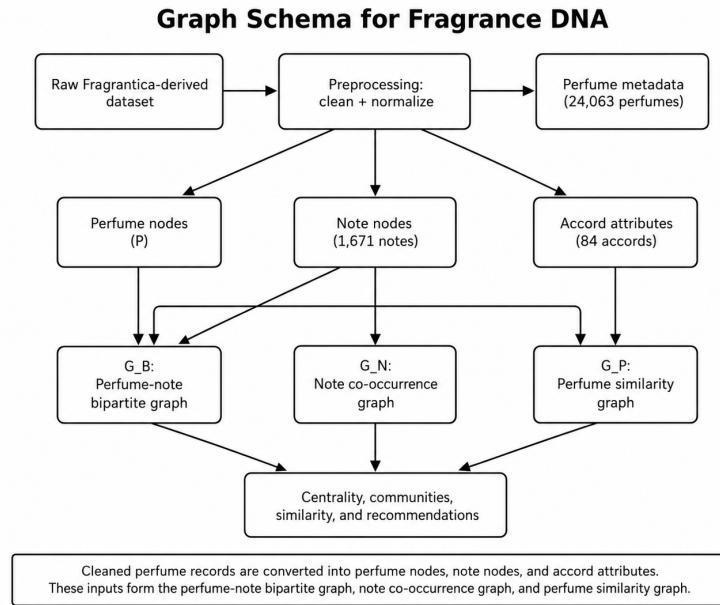


Figure 1: Graph construction pipeline for *Fragrance DNA*. The raw Fragrantica-derived dataset is cleaned into perfume nodes, note nodes, and accord attributes. These inputs are used to build the perfume-note bipartite graph, note co-occurrence graph, and perfume similarity graph.

recommendation. The graph-based recommendations are evaluated against popularity-based, rating-based, raw note-overlap, and same-brand baselines. This comparison tests whether graph structure adds value beyond simpler ranking strategies.

4.3 Network Analysis

Centrality results were generated for the note co-occurrence graph. The top notes by degree centrality include musk, sandalwood, bergamot, amber, vanilla, patchouli, cedar, jasmine, rose, and mandarin orange. These notes are highly connected in the note graph and are likely to influence both graph structure and perfume similarity.

The same set of broadly common fragrance notes also appears strongly in betweenness centrality and PageRank results. This suggests that notes such as musk, sandalwood, amber, vanilla, and bergamot are not only frequent, but also structurally important in connecting different scent profiles. In practical terms, these notes act as bridges across fragrance families: they appear in many perfumes and help connect otherwise different scent profiles through shared composition patterns.

4.4 Centrality Interpretation

The centrality results provide insight into which notes act as bridges across the fragrance space. Degree centrality identifies notes that co-occur with many other notes, while betweenness centrality highlights notes that lie on many shortest paths between other scent components. PageRank gives higher scores to notes that are connected to other important notes. When the same notes appear across these measures, they are likely to be structurally important rather than only frequent.

The top-ranked notes are mostly broad perfumery materials such as musk, sandalwood, bergamot, amber, vanilla, patchouli, cedar, jasmine, rose, and mandarin orange. This pattern is meaningful because these notes appear across many scent families. For example, bergamot and mandarin orange often connect citrus and fresh fragrances, while sandalwood, cedar, patchouli, amber, and vanilla connect woody, amber, and warm scent profiles. Floral notes such as jasmine and rose also bridge many fragrance styles because they appear in both traditional floral compositions and mixed modern perfumes.

This analysis supports the recommendation task because central notes help explain why perfumes from different brands or categories can still be structurally close in the graph. A perfume does not need to share every note with another perfume to be related; it may be connected through central notes that commonly bridge fragrance families.

4.5 Community Detection

Community detection was used to identify groups of notes that frequently appear together in the note co-occurrence graph. These communities help interpret the scent structure of the dataset beyond individual note frequency. Some communities are centered around broadly used perfumery materials, while others group notes that resemble woody, amber, floral, citrus, fruity, or gourmand scent families. Since scent descriptions are subjective and note labels may overlap across families, the communities are interpreted as graph-based clusters rather than fixed fragrance taxonomies.

The examples in Table 3 are interpretable as scent-profile clusters. Community 1 suggests a woody, amber, and resinous structure;

Table 3: Examples of detected note communities.

Comm.	Size	Interpretation
1	503	Woody/amber/resinous: bergamot, benzoin, patchouli, oud, vanilla
2	367	Fruity/floral: hibiscus, kiwi, pomegranate, nectarine
0	268	Common core: bergamot, musk, sandalwood, jasmine, patchouli

Community 2 groups fruit-forward and floral notes; and Community 0 contains common perfumery materials that act as a general scent-profile core. These results indicate that the note graph captures both specialized scent families and broadly shared ingredients used across many perfumes.

5 Evaluation

The evaluation compares the graph-based recommendation output against several simple baselines. Since the dataset does not contain user-level interaction histories, the evaluation focuses on item-level scent similarity rather than personalized user behavior. Recommended perfumes are evaluated using shared notes, note Jaccard similarity, shared accords, accord Jaccard similarity, same-brand percentage, average rating, and average rating count.

The evaluation is organized around three questions. First, we examine whether the constructed graphs have reasonable connectivity and interpretable central nodes. Second, we compare graph-based recommendations against simple non-graph baselines using shared-note and shared-accord metrics. Third, we inspect qualitative case studies to determine whether individual recommendations are explainable through scent-profile overlap.

5.1 Baselines

We compare the graph-based recommendations against four baselines. The **highest-rated baseline** recommends perfumes with the highest rating values, regardless of scent similarity. The **most-popular baseline** recommends perfumes with the largest number of ratings. The **same-brand baseline** recommends perfumes from the same brand as the query perfume, ranked by rating count and rating value. The **raw note-overlap baseline** recommends perfumes with the highest direct note overlap, measured using Jaccard similarity over perfume-note sets.

These baselines represent common non-graph strategies: recommending by rating, popularity, brand, or direct feature overlap. A useful graph-based method should outperform rating, popularity, and brand-only baselines on scent similarity while remaining competitive with raw note-overlap.

5.2 Baseline Comparison

For each query perfume in the top- k recommendation output, we compare the graph-based recommendations with recommendations produced by each baseline. Table 4 summarizes the comparison.

In Table 4, “Notes” and “Accords” refer to average shared notes and accords, while “Jac.” refers to Jaccard similarity and “Rating Ct.” refers to rating count.

The graph-based method performs much better than the highest-rated, most-popular, and same-brand baselines on scent similarity. It achieves an average of 5.88 shared notes and 2.80 shared accords per recommendation. In contrast, the highest-rated and most-popular baselines have fewer than 1.2 shared notes on average, showing that rating-based and popularity-based recommendation alone does not capture fragrance similarity.

The raw note-overlap baseline has the highest average shared-note score because it directly maximizes note overlap. However, the graph-based method remains close to this baseline while producing slightly stronger accord overlap. This suggests that the graph-based method captures broader scent-profile similarity through accords and graph relationships, rather than only matching direct note sets.

5.3 Qualitative Case Studies

We also inspected selected query perfumes from the top- k recommendation output. Table 5 summarizes three representative examples.

For *Accento Overdose Pride Edition* by Xerjoff, the top graph-based recommendation is *Accento Overdose*, also by Xerjoff. This recommendation shares 7 notes with the query perfume and has a note Jaccard similarity of 0.778, making it a strong same-line recommendation.

For *Classique Pride 2023* by Jean Paul Gaultier, the top recommendations include *Le Male Pride Edition* and *Le Male Pride 2023*. These recommendations share all 6 notes with the query perfume, giving a note Jaccard similarity of 1.000. They also share several accords, including citrus, white floral, sweet, fresh spicy, and musky. This case shows that the graph-based method can identify near-duplicate or closely related flankers within the same fragrance line.

For *Pride Edition Man* by Bruno Banani, the top recommendation is *Bruno Banani No Limits Man*, which shares 6 notes with the query perfume and has a note Jaccard similarity of 0.600. Other recommendations include perfumes from different brands, showing that the graph-based method can also identify cross-brand scent similarities.

5.4 Evaluation Discussion

Overall, the results show that the graph-based recommendation method produces meaningful and explainable recommendations. It performs substantially better than rating-based, popularity-based, and same-brand baselines on scent similarity, and it remains competitive with raw note-overlap.

The graph-based method is useful because it connects recommendation quality to interpretable graph evidence. Shared notes and accords explain direct scent similarity, while graph structure captures broader relationships among perfumes and scent components. This makes the recommendation output more transparent than ranking perfumes only by rating, popularity, or brand.

5.5 Interpretation of Results

The evaluation results show two important patterns. First, recommendation quality cannot be inferred from rating or popularity

Table 4: Comparison of graph-based recommendations against simple baselines.

Method	Notes	Note Jac.	Accords	Accord Jac.	Brand %	Rating	Rating Ct.
Graph-based	5.88	0.583	2.80	0.445	60.0	3.52	1233.2
Highest-rated	0.96	0.055	1.32	0.168	0.0	4.85	28.2
Most-popular	1.16	0.055	1.28	0.166	0.0	3.83	29510.6
Same-brand	1.76	0.106	1.36	0.184	100.0	4.14	9397.6
Raw note-overlap	6.20	0.612	2.60	0.408	68.0	3.35	1080.0

Table 5: Example graph-based recommendations.

Query perfume	Query brand	Recommended perfume	Recommended brand	Shared notes	Note Jac.
Accento Overdose Pride Edition	Xerjoff	Accento Overdose	Xerjoff	7	0.778
Classique Pride 2023	Jean Paul Gaultier	Le Male Pride Edition	Jean Paul Gaultier	6	1.000
Pride Edition Man	Bruno Banani	Bruno Banani No Limits Man	Bruno Banani	6	0.600

alone. The highest-rated and most-popular baselines achieve strong rating-oriented scores, but they share very few notes or accords with the query perfumes. This shows that popularity signals are useful for identifying well-known or well-liked perfumes, but they do not necessarily represent scent similarity.

Second, the graph-based method remains close to raw note-overlap while improving accord overlap. This is important because raw note-overlap is optimized directly for shared notes, whereas the graph-based method combines note similarity, accord similarity, graph proximity, and a popularity prior. As a result, the graph-based recommendations are not only similar by direct note sets, but also supported by broader graph structure and scent-profile evidence.

The qualitative examples support this interpretation. Same-line recommendations such as *Accento Overdose Pride Edition* to *Accento Overdose* show that the method can identify near variants. The *Classique Pride 2023* example shows that the system can detect closely related flankers with identical note overlap. The *Pride Edition Man* example shows that the method can still produce meaningful recommendations when the note overlap is lower, which is useful for finding perfumes that are related but not exact duplicates.

6 Visualization and Interpretation

Figure 2 compares raw note frequency with graph-based note centrality. The frequency chart shows which notes appear most often in the processed perfume-note edge list, while the centrality chart shows which notes are most connected in the note co-occurrence graph.

Comparing these two views helps separate simple popularity from graph-based structural importance. For example, musk, bergamot, vanilla, sandalwood, amber, and jasmine appear prominently in both views, suggesting that they are both common scent components and important connectors in the graph.

Because the full perfume similarity graph contains over one million edges, full-graph visualization would be difficult to interpret. Instead, the project uses filtered views such as centrality rankings, frequency charts, and selected recommendation examples. These

summaries connect the graph results back to the project goal of explainable fragrance recommendation.

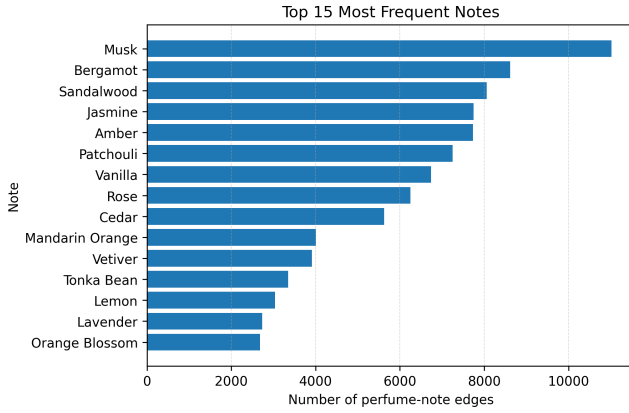
7 Limitations

The main limitation of the evaluation is the absence of user-level interaction histories. Since the dataset does not include user clicks, purchases, reviews, or personalized preference histories, the evaluation measures item-level scent similarity rather than personalized recommendation accuracy. Therefore, the results show that the system recommends perfumes with similar scent profiles, but they do not directly prove that individual users would prefer those recommendations.

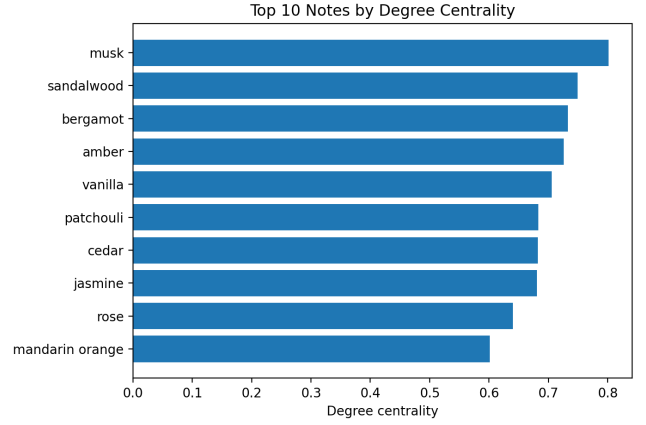
There are also dataset limitations. Fragrance descriptions are subjective, and note or accord labels may be inconsistent because they come from user-facing fragrance records. Some records are missing release year or perfumer information. Notes also vary in specificity: broad notes such as musk or amber may appear in many perfumes, while rare notes may create very specific but less stable similarity links.

Finally, visualization is challenging because the constructed graphs are large. The perfume similarity graph contains over one million edges, so full-graph visualization would be difficult to interpret. For this reason, the report focuses on summary statistics, centrality rankings, selected recommendation examples, and interpretable graph-based comparisons.

A further limitation is that the final results depend on design choices such as the perfume similarity threshold, the weighting of the composite recommendation score, and the use of Jaccard-based note and accord overlap. Different thresholds or weights could change the density of the perfume similarity graph and the order of top- k recommendations. A more complete robustness analysis would compare several threshold values and score-weight settings to test whether the same central notes, communities, and recommendation patterns remain stable.



(a) Top 15 notes by frequency.



(b) Top notes by degree centrality.

Figure 2: Frequency-based and graph-based views of important fragrance notes. The frequency chart shows which notes occur most often, while the centrality chart shows which notes are most connected in the note co-occurrence graph.

8 Conclusion

This project presented *Fragrance DNA*, a graph-based approach for analyzing perfume scent profiles and generating explainable fragrance recommendations. Starting from a FrAGRANTICA-derived dataset, we constructed perfume-note, note co-occurrence, and perfume similarity graphs. These graph structures made it possible to identify central fragrance notes, examine note communities, and recommend perfumes using scent-profile similarity rather than only ratings, popularity, or brand.

The results show that common notes such as musk, sandalwood, bergamot, amber, vanilla, patchouli, cedar, jasmine, rose, and mandarin orange act as important connector nodes in the note co-occurrence graph. The recommendation evaluation also shows that the graph-based method substantially outperforms rating-based, popularity-based, and same-brand baselines on shared-note and shared-accord similarity. Although raw note-overlap remains slightly stronger on direct note Jaccard similarity, the graph-based method remains competitive while providing broader interpretability through graph structure and accord similarity.

The main limitation is that the dataset does not include user-level interaction histories, so the evaluation measures item-level scent similarity rather than personalized user satisfaction. Future work could incorporate user reviews, ratings, seasonality, gender labels, or richer natural-language descriptions to improve recommendation quality and personalization.

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